

7.5 Factoring Trinomials

Lesson Introduction

 Benchmark	MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.		 Learning Target	I can factor trinomials.
 Benchmark Coherence	Building On MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.		Working Toward MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real or complex number system.	
 Essential Question	How can you apply what you've learned to factor trinomials?			
 Lesson Narrative	In this lesson, students continue their learning with factoring trinomials. Students will engage in a collaborative Exploration relating the terms within given expressions to the factors of the expressions. They will then engage in a culminating card sort task where they will match a quadratic expression with its representations using algebra tiles, area model, expanded form, and factored form. Additionally, students will learn to factor trinomials when the leading coefficient has multiple factors before wrapping up the lesson.			
 Component Summary	7.5.1 Warm-Up (~5 min.)	Problem-solving activity about walking the state of Florida (SE p. 382)	7.5.3 Activity (15-20 min.)	Card sort activity matching quadratic expression, algebra tiles, area model, expanded form, and factored form (SE p. 385)
	7.5.2 Exploration (10 min.)	Collaborative exploration relating terms in trinomials to factors of the expression (SE p. 383)	7.5.4 Guided Instruction (10 min.)	Formal instruction on factoring trinomials when the leading coefficient has multiple factors (SE p. 385)
	7.5.5 Wrap-Up (~5 min.)	Reflect on the lesson using a KWL chart (SE p. 386)		
 Resources	Glossary		Additional Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		For pacing purposes, consider completing the card sort activity on your own to determine the amount of time you want to provide for this activity.	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may attempt to memorize a list of steps to factor, solidifying misconceptions that may include applying methods of factoring that work when $a = 1$ to expressions in which $a \neq 1$.	Students are applying what they've learned in the prior lessons in a culminating experience in the 7.5.3 Activity. Consider the best way to group students for this activity based on their performances in the prior lessons, including whether there is a group of students that would benefit from this activity being more guided.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns	MA.K12.MTR.1.1: Participate in Effortful Learning
	7.5.3 Activity has students working on a card sort where they match the quadratic expression to its representation with algebra tiles, area model, expanded form, and factored form. Students can use this instructional routine to take turns in the work of this activity by spotting matches and justifying their answers to their group.	7.5.3 Activity requires students work with partners or small groups to complete a card sort. Students will need to analyze the problem in a way that makes sense given the task. They will also need to stay engaged and maintain a positive mindset when working to solve the task in this activity. Consider ways to recognize students' efforts when solving challenging problems.
 Differentiate Instruction	For 7.5.2 Exploration, students will work with a visual diagram that represents the use of algebra tiles. Consider providing physical algebra tiles or paper cut-out representations to allow for a hands-on Exploration.	
	7.5.3 Activity is a collaborative card sort that can be done with partners or small groups. Consider grouping students based on their performances in the prior factoring lessons. If data is available, consider creating groups based on their strengths in each method for a dynamic learning experience for all the students in that group.	
Lesson Notes:		

7.5.1 Warm-Up: Walking Florida

Component Overview: Students will be presented with some data about the state of Florida. They will use this data to answer questions to apply mathematical knowledge to solve a problem.



7.5.1 Warm-Up: Walking Florida

1. Florida has the longest coastline, 1197 miles, in the contiguous United States. The distance from Key West to the farthest west Florida border on I-10 is 839 miles. The average person walks 3 miles per hour (mph).

- a. How many hours would it take someone walking 3 mph to walk from Key West to Florida's western border?

$$839 \div 3 = 279.\overline{666}$$

It would take about 280 hours.

- b. How much longer, in days, would it take someone to walk along Florida's coastline than to walk from Key West to Florida's western border?

$$(1197\text{mi} - 839\text{mi}) = 358\text{mi}$$

$$358\text{mi} \cdot \frac{1\text{ hour}}{3\text{mi}} \cdot \frac{1\text{ day}}{24\text{ hours}} = 4.97\overline{2}\text{ days}$$

It would take approximately 5 more days for someone to walk along the entire coastline than for them to just walk from Key West to Florida's western border.



Consider employing a Think-Pair-Share routine for this activity. Allow students a few minutes to try this on their own. Then have them work with a partner to share their work before coming to a consensus on the correct answer prior to reviewing the answer as a whole group.

7.5.2 Exploration: Relating Terms and Factors

Component Overview: Students will engage in a collaborative Exploration relating terms and factors. In this activity, students will determine the quadratic expression that results after the multiplication of two binomials. Students will then practice using algebra tiles to factor trinomials. This activity is designed to be completed with a partner or small group.



7.5.2 Exploration: Relating Terms and Factors

- The steps for using the distributive property to multiply $(2x + 3)(2x - 2)$ are shown.

Step	$(2x + 3)(2x - 2)$	Result
Distribute $2x$ to $2x$	$(2x + 3)(2x - 2)$ 	$4x^2$
Distribute $2x$ to -2	$(2x + 3)(2x - 2)$ 	$-4x$
Distribute 3 to $2x$	$(2x + 3)(2x - 2)$ 	$6x$
Distribute 3 to -2	$(2x + 3)(2x - 2)$ 	-6

- Based on the results in the table, what quadratic expression is the product of $(2x + 3)(2x - 2)$?
 $4x^2 + 2x - 6$
- Tyra and Min-Joon were discussing how to find the x -term of the quadratic expression for $(2x + 3)(2x - 2)$.

Tyra	Min-Joon
The x -term of $(2x + 3)(2x - 2)$ can be quickly found by adding 3 and -2 .	Looking at the table, the x -term of the quadratic expression is $2x$. I think you could just use what repeats.

Discuss with your partner both students' strategies.
Answers will vary.

- Both students have made a mistake. Describe to Tyra and Min-Joon how to find the x -term of the quadratic expression for $(2x + 3)(2x - 2)$.
Distribute each term in the first binomial with each term in the second. Then, combine like terms.



Use student responses from Lesson 4 Wrap-Up or this lesson's Warm-Up. Strategically group students for both this activity and the Card Sort.



What do you notice about the methods Tyra and Min-Joon used and what was found in the table you just completed?



Consider having students actually try the original methods that Tyra and Min-Joon suggested to see what answers would have been produced. Use this as a springboard for conversation around the errors and solutions identified in part c.

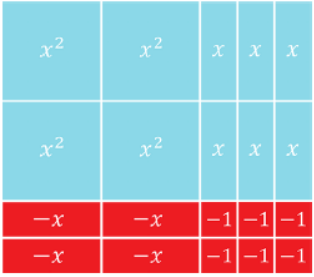
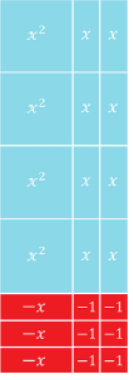
7.5.2 Exploration: Relating Terms and Factors (continued)

- d. Discuss with your partner whether or not the x -term of the quadratic expression $(4x - 3)(x - 2)$ is different from the x -term of the quadratic expression $(2x - 3)(2x - 2)$. Summarize your discussion.

No, the first trinomial is $4x^2 - 11x + 6$, whereas the second is $4x^2 - 10x + 6$, so although the first and third terms are the same, the middle terms are not.

2. Consider the trinomials $4x^2 + 2x - 6$ and $4x^2 + 5x - 6$.

- a. Use the algebra tiles to factor each trinomial.

Trinomial	$4x^2 + 2x - 6$	$4x^2 + 5x - 6$
Algebra Tiles		
Factored Form	$(2x - 2)(2x + 3)$ <i>or $2(x - 1)(2x + 3)$</i>	$(4x - 3)(x + 2)$

- b. The trinomials have the same leading term, $4x^2$. List the two ways $4x^2$ is factored.
 $x, 4x$ and $2x, 2x$
- c. List the factor pairs of -24 .
 $1, -24; 2, -12; 3, -8; 4, -6; -1, 24; -2, 12; -3, 8; -4, 6$
- d. Which pair of factors of -24 add to 2?
 $-4, 6$



What terms are being multiplied to find the x -term? How can knowing that help you quickly determine if the x -terms will be different?



If available, consider providing students with physical algebra tiles to use when answering question 2. If they are not available, consider creating paper cut-out versions of what's needed for these trinomials to allow for a hands-on experience.



What do you notice is different about the first trinomial's factored form?



What are some of the challenges when working with factor pairs of a negative number?

7.5.2 Exploration: Relating Terms and Factors (continued)

- e. Discuss with your partner how the pair of factors of -24 that add to 2 is related to the factored form of $4x^2 + 2x - 6$. Show the factors along the algebra tiles.

Answers will vary. $2x$ is the sum of the products of the innermost and outermost terms in the binomial factors.

	x	x	1	1	1
x	x^2	x^2	x	x	x
x	x^2	x^2	x	x	x
-1	$-x$	$-x$	-1	-1	-1
-1	$-x$	$-x$	-1	-1	-1



Create a space for whole-group discussion of part e as it relates to what they've done in question 2. The knowledge from this Exploration will assist them in their work in the upcoming card sort activity.

7.5.3 Activity: Card Sort

Component Overview: Students will work in partner pairs or small groups to complete a card sort activity. In this activity, students will match the card that has a quadratic expression on it with its corresponding representation using algebra tiles, area model, expanded form, and factored form.



7.5.3 Activity: Card Sort

1. Work with your partner to sort the cards. Write the letter of the matching card for each representation.

Quadratic Expression	Algebra Tiles	Area Model	Expanded Form	Factored Form
$6x^2 - x - 2$	<i>P</i>	<i>D</i>	<i>T</i>	<i>F</i>
$6x^2 + x - 2$	<i>K</i>	<i>O</i>	<i>Z</i>	<i>B</i>
$6x^2 - 7x + 2$	<i>E</i>	<i>U</i>	<i>Y</i>	<i>G</i>
$6x^2 - 11x - 2$	<i>R</i>	<i>J</i>	<i>A</i>	<i>M</i>
$6x^2 + 11x - 2$	<i>V</i>	<i>S</i>	<i>L</i>	<i>H</i>
$6x^2 - 11x + 2$	<i>N</i>	<i>I</i>	<i>W</i>	<i>C</i>

2. Describe one strategy you and your partner used to sort the cards.

Answers will vary. My partner and I started with the tiles first because we are more comfortable with that. My partner and I started with the expanded form because that was the easiest card to eliminate from the stack.



Pre-cut and group cards into plastic bags with a group number on the outside to pass out to each group. To reduce loss of instructional time, consider having students work with the same partner or small group that they worked with in the previous component.

How students respond to each matching category in this activity can provide insight into misconceptions and areas that need additional support prior to the end of the unit.

Consider creating an index card or printout with the correct letters for each row. Keep this with you as you circulate the room to better provide just-in-time feedback to students.



Teachers may have algebra tiles on hand to help students manipulate tiles or use a virtual set of algebra tiles. The teacher may wish to modify the activity if time is a concern. The best modification may be to give students a few of the correct matches. Other modifications might take away from the integrity of students making their own connections.

7.5.4 Guided Instruction: Factoring Trinomials in the Form $ax^2 + bx + c$ when a Has Multiple Factors

Component Overview: Students will receive guided instruction on how to factor trinomials when the leading coefficient has multiple factors.



7.5.4 Guided Instruction: Factoring Trinomials in the Form $ax^2 + bx + c$ when a has Multiple Factors

Factoring trinomials of the form $ax^2 + bx + c$, where the value of a has multiple factors, requires all of the factors of a to be considered when factoring using algebra tiles or the area model.

1. Complete the table for $10x^2 + x - 3$.

	<u>$2x$</u>	<u>-1</u>
<u>$5x$</u>	$10x^2$	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $10x$ ☐ $-10x$ ☐ $3x$ ☐ $-3x$ ☐ $15x$ ☐ $-15x$ ☐ $5x$ ☐ $-5x$ ☐ $30x$ ☐ $-30x$
<u>3</u>	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $10x$ ☐ $-10x$ ☐ $3x$ ☐ $-3x$ ☐ $15x$ ☐ $-15x$ ☐ $5x$ ☐ $-5x$ ☐ $30x$ ☐ $-30x$	-3

2. Which expanded form of $10x^2 + x - 3$ could be used to factor the expression by grouping?

- ☐ $10x^2 + 5x - 6x - 3$
- ☐ $10x^2 - 5x + 6x - 3$
- ☐ $10x^2 - 10x + 3x - 3$
- ☐ $10x^2 + 10x - 3x - 3$

3. Use the expanded form to factor $10x^2 + x - 3$.
 $(2x - 1)(5x + 3)$



Consider the placement of this activity in the lesson. Students will have just finished a collaborative and potentially lively discussion during the card sort activity.

To maintain instructional momentum, consider allowing students to remain with their partner or small group and use this material in a collaborative release structure. For example, read the narrative prior to question 1 and release students to try question 1 with their partner or group and then review as a whole group.

Repeat this process for questions 2 and 3 or allow students to independently practice on their own while pulling a small group for accelerated learning.

7.5.5 Wrap-Up: Reflection

Component Overview: Students will complete a reflection activity regarding the 7.5.2 Exploration activity. This activity is designed to be done independently to collect qualitative student data on their experiences with the 7.5.2 Exploration activity.



7.5.5 Wrap-Up: Reflection

1. Complete the table.

Answers will vary.

The terms and information used in the exploration and activity that I knew were ...	Something I wondered during the exploration or activity was ...	Something I learned during the exploration or activity was ...



Consider pairing students to discuss their reflections to see if peers can answer any of their questions or wonderings.